Universidade de São Paulo

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Projeto 5

Display de 7 segmentos

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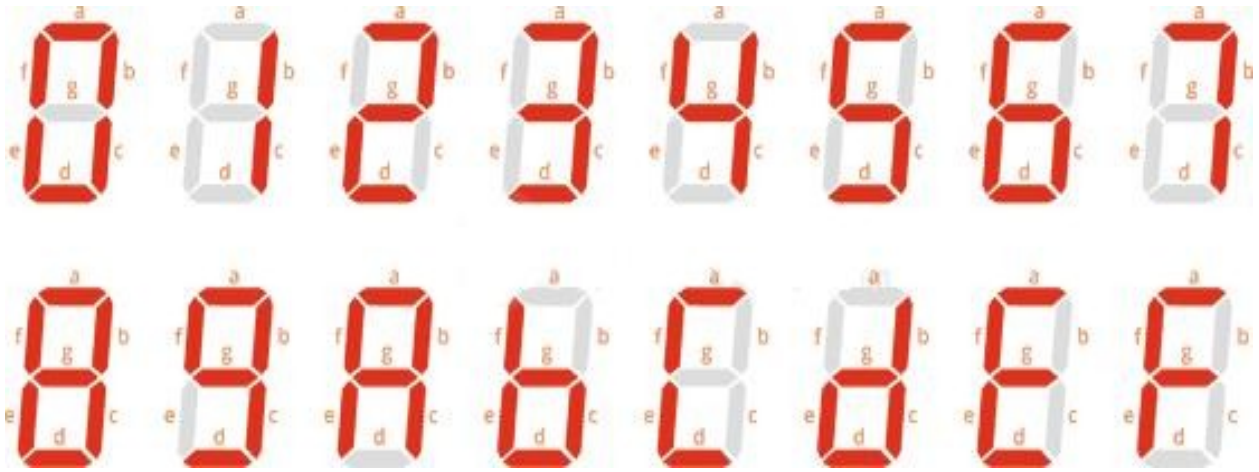
Objetivo do trabalho:

- Projetar o controle de um display de 7 segmentos (mostrando em hexadecimal) para entradas (4bits).

Desenvolvimento do projeto:

De acordo com o manual da placa Altera DE0-CV será tratado como apagado o devido segmento quando este estiver em nível alto (1) e aceso quando estiver em nível baixo (0) e para representar cada segmento serão escolhidas as letras “a, b, c, d, e, f, g” como mostrado na figura abaixo.



Para representar as entradas de 4 bits serão escolhidos os termos “b3, b2, b1, b0”, onde “b0” será o binário menos significativo e “b3” o binário mais significativo.

Logo abaixo está representada a Tabela-Verdade, com as entradas (b3,b2,b1,b0), as saídas (a, b, c, d, e, f, g) e os hexadecimais (Hx) que serão representados no display.

Hx	b3	b2	b1	b0	a	b	c	d	e	f	g
1	0	0	0	0	0	0	0	0	0	0	1
E	0	0	0	1	1	0	0	1	1	1	1
2	0	0	1	0	0	0	1	0	0	1	0
3	0	0	1	1	0	0	0	0	1	1	0
4	0	1	0	0	1	0	0	1	1	0	0
5	0	1	0	1	0	1	0	0	1	0	0
6	0	1	1	0	0	1	0	0	0	0	0
F	0	1	1	1	0	0	0	1	1	1	1
8	1	0	0	0	0	0	0	0	0	0	0
9	1	0	0	1	0	0	0	0	1	0	0
A	1	0	1	0	0	0	0	1	0	0	0
b	1	0	1	1	1	1	0	0	0	0	0
H	1	1	0	0	0	1	1	0	0	0	1
d	1	1	0	1	1	0	0	0	0	1	0
U	1	1	1	0	0	1	1	0	0	0	0
0	1	1	1	1	0	1	1	1	0	0	0

A seguir, a partir da Tabela-Verdade, serão montados os mapas de Karnaugh para se obter as expressões lógicas.

- Mapa de "a" :

b1 b0 \ b3 b2	00	01	11	10
00	0	1	0	0
01	1	0	1	0
11	0	0	0	1
10	0	0	0	0

Expressão Simplificada:

a = /(b3+b1).(b2⊕b0) + b3.b0.(b2⊕b1)

- Mapa de "b" :

b1 b0 \ b3 b2	00	01	11	10
00	0	0	1	0
01	0	1	0	0
11	0	0	1	1
10	0	1	1	0

Expressão Simplificada:

b = b2.[(b3+b1)⊕b0] + b3.b1.b0

- Mapa de “c” :

b1 b0 \ b3 b2	00	01	11	10
00	0	0	1	0
01	0	0	0	0
11	0	0	1	0
10	1	0	1	0

Expressão Simplificada:

c = /(b3+b2+b0).b1 + b3.b2.(b1+/b0)

- Mapa de “d” :

b1 b0 \ b3 b2	00	01	11	10
00	0	1	0	0
01	1	0	0	0
11	0	1	1	0
10	0	0	0	1

Expressão Simplificada:

d = /(b3+b1).(b2⊕b0) + b1.[/(b2+b0).b3 + b2.b0]

- Mapa de “e” :

b1 b0 \ b3 b2	00	01	11	10
00	0	1	0	0
01	1	1	0	1
11	1	1	0	0
10	0	0	0	0

Expressão Simplificada:

<u>e = /(b3+b1).b2 + b0.[/b3 + /(b2+b1)]</u>

- Mapa de “f” :

b1 b0 \ b3 b2	00	01	11	10
00	0	0	0	0
01	1	0	1	0
11	1	1	0	0
10	1	0	0	0

Expressão Simplificada:

<u>f = /(b3+b2).(b1+b0) + b0.(/b3.b1+b3.b2./b1)</u>

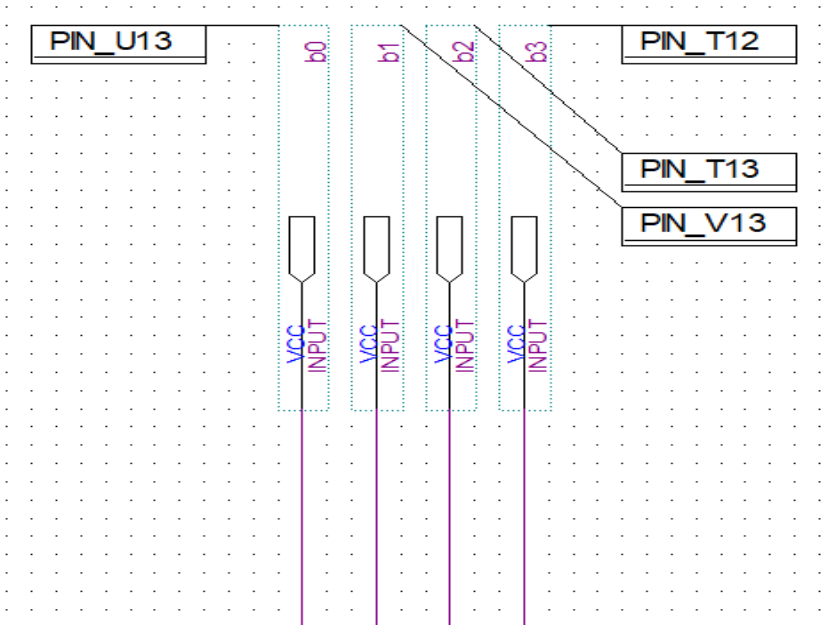
- Mapa de “g” :

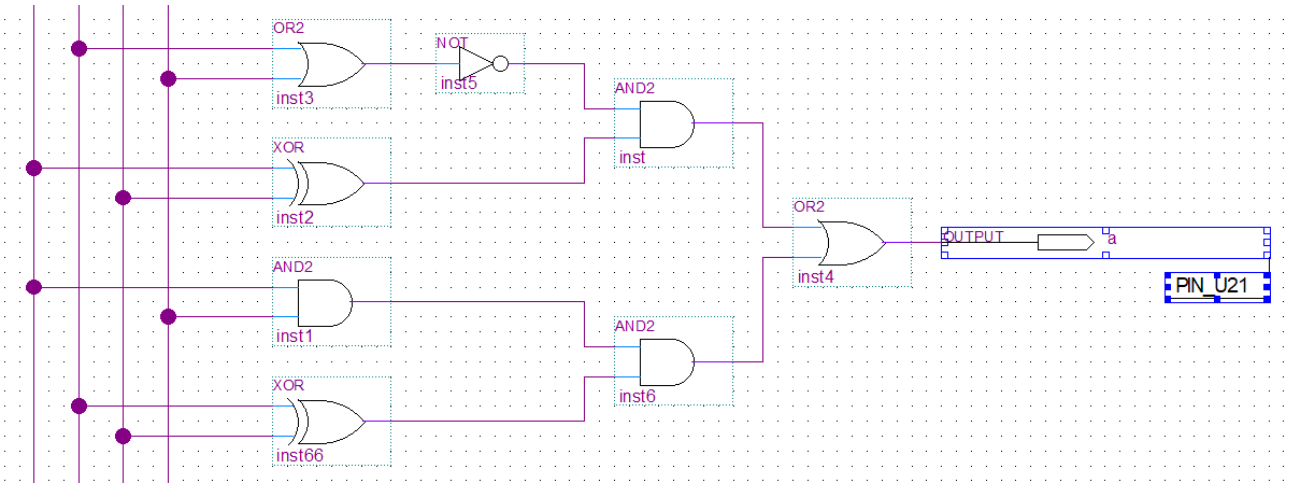
b1 b0 \ b3 b2	00	01	11	10
00	1	0	1	0
01	1	0	0	0
11	0	1	0	0
10	0	0	0	0

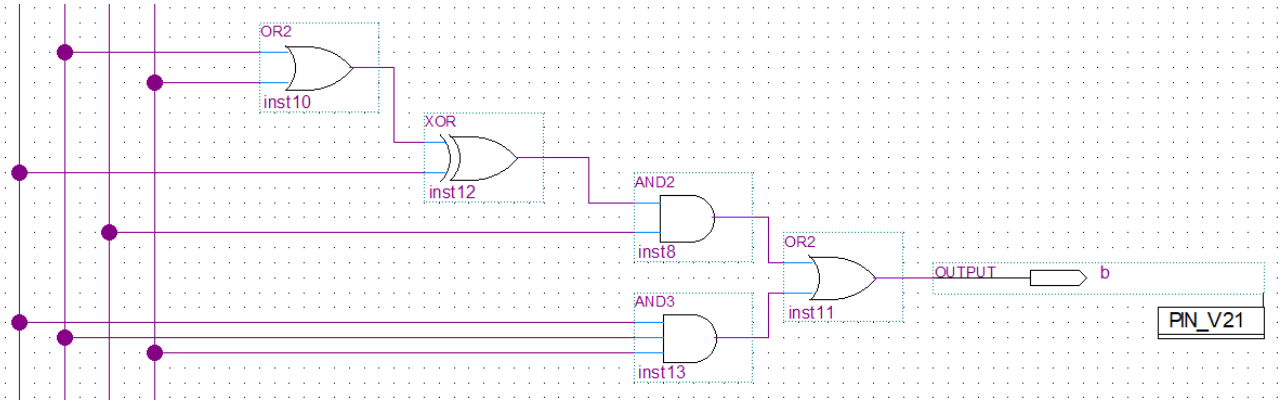
Expressão Simplificada:

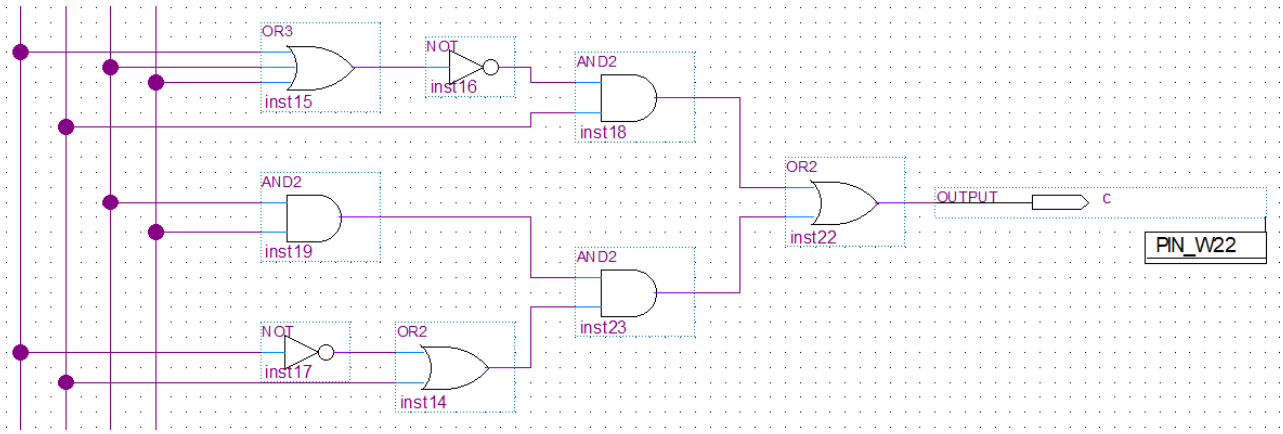
g = b2.[b3./(b1+b0)+/b3.b1.b0] + /(b3+b2+b1)

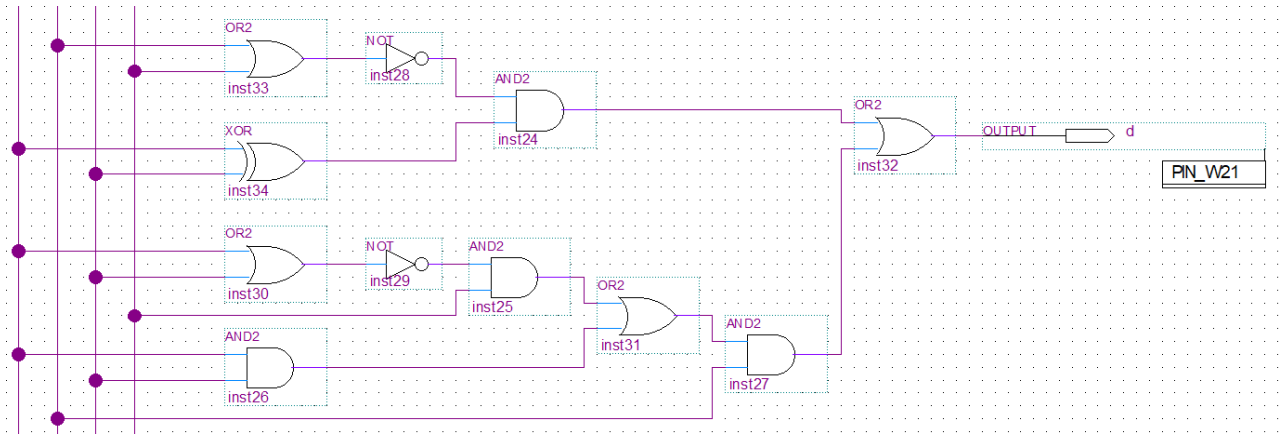
A partir da expressões lógicas simplificadas, serão montados os circuitos no *Quartus II*.

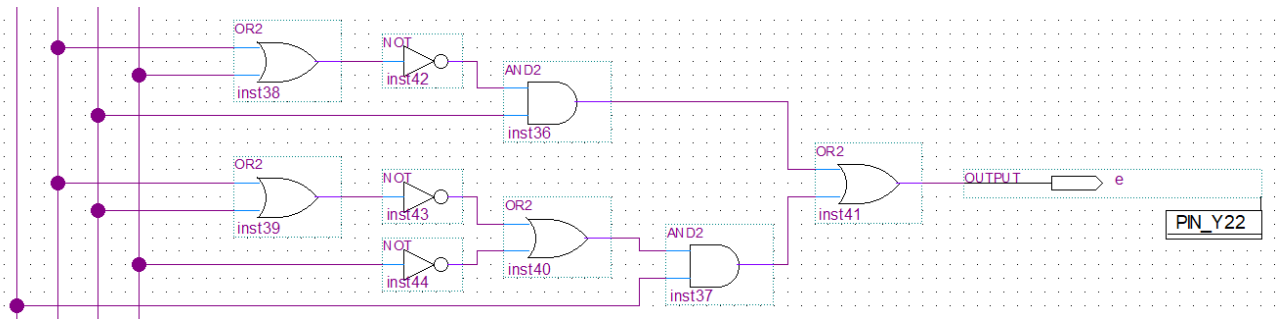


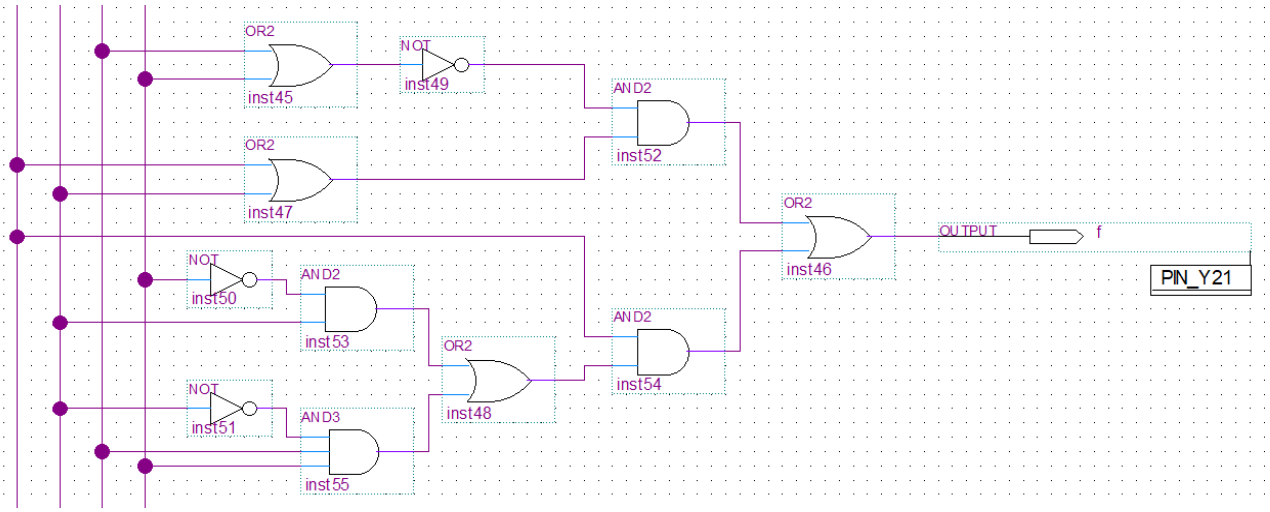

OR2
inst3
NOT
inst5
AND2
inst
XOR
inst2
OR2
inst4
OUTPUT
a
PIN_U21
AND2
inst1
AND2
inst6
XOR
inst66

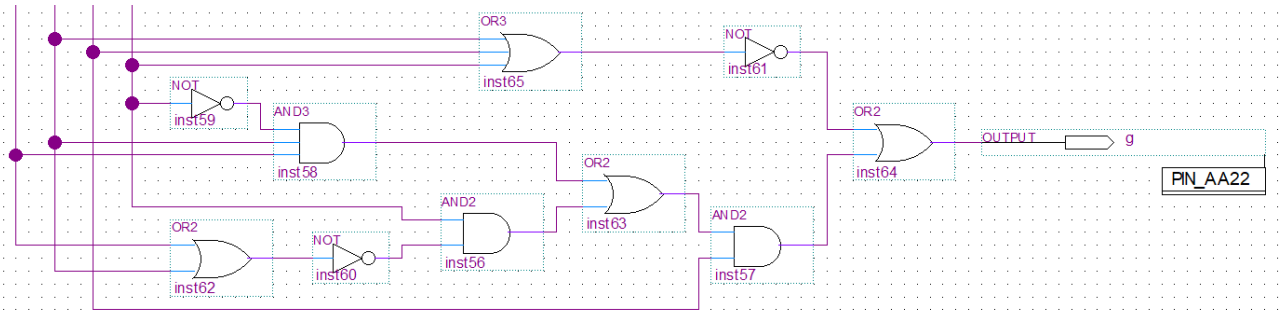

OR2
inst10
XOR
inst12
AND2
inst8
OR2
inst11
OUTPUT
b
PIN_V21
AND3
inst13


OR3
inst15
NOT
inst16
AND2
inst18
OR2
inst22
OUTPUT
c
PIN_W22
AND2
inst19
AND2
inst23
NOT
inst17
OR2
inst14


OR2
inst33
NOT
inst28
AND2
inst24
OR2
inst32
OUTPUT
d
PIN_W21
XOR
inst34
OR2
inst30
NOT
inst29
AND2
inst25
OR2
inst31
AND2
inst27
AND2
inst26







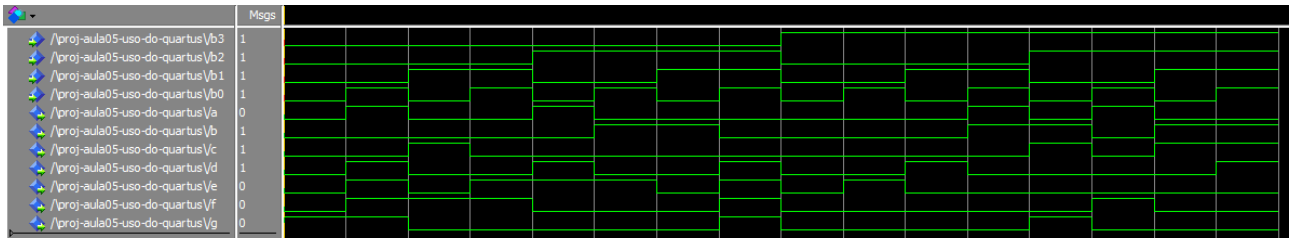
As seguintes pinagens foram feitas utilizando *Cyclone V* para a placa *DEV0-CV:*

	Status	From	To	Assignment Name	Value	Enabled	Entity	Comment	Tag
1	Ok		in b0	Location	PIN_U13	Yes			
2	Ok		in b1	Location	PIN_V13	Yes			
3	Ok		in b2	Location	PIN_T13	Yes			
4	Ok		in b3	Location	PIN_T12	Yes			
5	Ok		out a	Location	PIN_U21	Yes			
6	Ok		out b	Location	PIN_V21	Yes			
7	Ok		out c	Location	PIN_W22	Yes			
8	Ok		out d	Location	PIN_W21	Yes			
9	Ok		out e	Location	PIN_Y22	Yes			
10	Ok		out f	Location	PIN_Y21	Yes			
11	Ok		out g	Location	PIN_AA22	Yes			

Após ser compilado com sucesso:

Flow Summary	
Flow Status	Successful - Wed Jun 16 00:07:35 2021
Quartus II 64-Bit Version	14.0.0 Build 200 06/17/2014 SJ Web Edition
Revision Name	proj-aula05-uso-do-quartus
Top-level Entity Name	proj-aula05-uso-do-quartus
Family	Cyclone V
Device	5CEBA4F23C7
Timing Models	Final
Logic utilization (in ALMs)	4 / 18,480 (< 1 %)
Total registers	0
Total pins	11 / 224 (5 %)
Total virtual pins	0
Total block memory bits	0 / 3,153,920 (0 %)
Total DSP Blocks	0 / 66 (0 %)
Total HSSI RX PCSs	0
Total HSSI PMA RX Deserializers	0
Total HSSI TX PCSs	0
Total HSSI PMA TX Serializers	0
Total PLLs	0 / 4 (0 %)
Total DLLs	0 / 4 (0 %)

Foi compilado no *ModelSim,* sendo apresentado a seguir:



Portanto, da simulação, confere-se que está coerente com a Tabela-Verdade.